COVID-19 GLOBAL IMPACT DASHBOARD

A data analytics project using Python, Pandas and Plotly

Project overview
This project explores publicly available historical COVID-19 data for India, the United States, Brazil and the United Kingdom. It cleans and organizes country-level time-series records, calculates comparison metrics, and presents trends through interactive visualizations.

1. Objectives

- Analyze reported COVID-19 cases, deaths and vaccination trends over time.
- Compare selected countries using population-adjusted measures where available.
- Calculate a seven-day rolling average of newly reported cases.
- Communicate the results with clear charts and a concise executive summary.

2. Tools and Technologies

Tool	Purpose
Python	Project programming and analysis
Pandas / NumPy	Data loading, cleaning, filtering and calculations
Plotly Express	Interactive time-series charts
Google Colab	Cloud notebook environment
ReportLab	Generation of the executive summary PDF

3. Dataset

Source: Our World in Data COVID-19 dataset. The notebook loads the public CSV data and filters records for India, United States, Brazil and United Kingdom. Fields used include location, date, population, total cases, new cases, total deaths and vaccination information, depending on availability in the source dataset.

4. Analysis and Visualizations

- **Cases per million:** compares cumulative reported cases relative to population.
- **Deaths per million:** compares cumulative reported deaths relative to population.
- **Vaccination coverage:** tracks the percentage of the population fully vaccinated, where data is available.
- **Seven-day average:** smooths daily new-case counts to make trends easier to interpret.

5. Project Workflow

Step	Activity
1	Load the public COVID-19 CSV dataset into Google Colab.
2	Inspect columns and filter the four selected countries.

Step	Activity
3	Convert dates and numeric fields; handle missing values used in calculations.
4	Calculate population-adjusted metrics and seven-day rolling averages.
5	Build interactive Plotly charts and compare country trends.
6	Export the cleaned dataset and executive summary PDF.

6. Key Insights to Report

Use the actual output from your notebook to write three to five findings. Do not guess or copy example statistics. Suggested prompts:

- Which selected country has the highest cumulative cases in the latest available record?
- How do deaths per million differ between the selected countries?
- Which country has the highest recorded full-vaccination percentage in the available data?
- When do the seven-day average case curves rise or fall most noticeably?

7. Files Produced

File	Description
COVID_Cleaned_Data.csv	Filtered and processed country-level data.
COVID_19_Executive_Summary.pdf	One-page summary with country comparison.
COVID-19_Global_Impact_Dashboard.ipynb	Google Colab notebook containing the code, outputs and charts.

8. How to Run

- Open the notebook in Google Colab.
- Run the cells from top to bottom with an active internet connection.
- The notebook downloads the public dataset, processes the data and displays interactive charts.
- Run the export cell to download the cleaned CSV and executive summary PDF.

9. Limitations and Data Notes

COVID-19 reporting practices and completeness vary by country and date. Missing values do not necessarily mean zero cases or zero vaccinations. Population-adjusted indicators improve comparability but do not account for differences in testing, reporting, age structure or healthcare access. The dataset contains historical reported figures; it should not be treated as a live count of active infections. The latest date available can differ by country and by indicator.

Data source: Our World in Data — COVID-19 dataset (https://ourworldindata.org/coronavirus).

This README documents the project workflow and deliverables. Add your own verified numerical findings from the notebook before final submission.